Container Optimization in HPC: Benchmarking Bare-Metal vs. Container Performance

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Background and Motivation 1

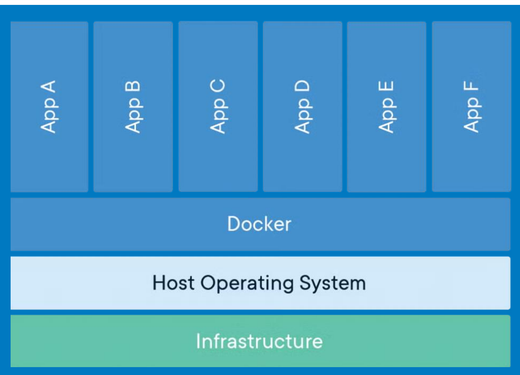


Figure 1 Docker Container Architecture Stack[1]

- Containers make software portable across machines but add an extra layer between the application and hardware.

Containers on Mac 2

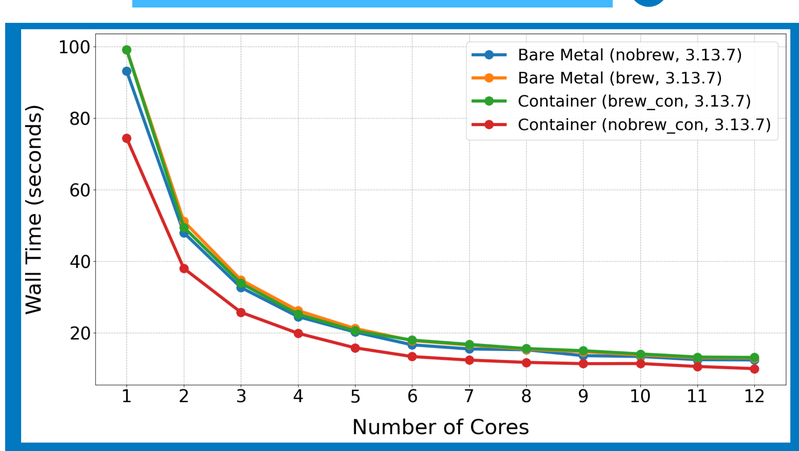


Figure 2 CPU mpi4py 100K-Operation Strong-Scaling Runtime (Python 3.13.7)

- Condensed container out performed bare metal.
- No container-specific penalty was observed once a condensed build was used
- Results of container outperforming BareMetal were consistent on both HPC and local environments.

Problem

- Researchers use containers to keep software environments consistent across different systems.
- Unclear how much performance is lost when using containers instead of bare metal.

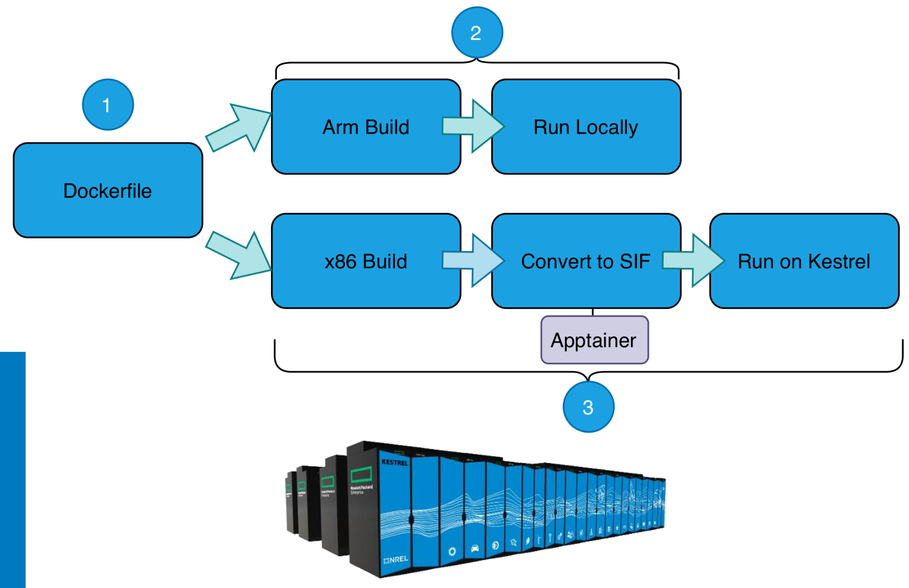


Figure 2 NLR Cluster Kestrel

Future Work

- Containers can perform better than BareMetal depending on configurations
- Expand benchmarking to GPU nodes and evaluate whether containerized GPU workflows match native performance.

References

1. Docker, Inc. (n.d.). What is a container? https://www.docker.com/resources/what-container/
2. Panda, D. K., Subramoni, H., Chu, C., & Bayatpour, M. (2025, December 5). MVAPICH: MPI over InfiniBand, Omni-Path, Ethernet/iWARP, RoCE, and Slingshot – OSU Micro-Benchmarks. Network-Based Computing Laboratory, The Ohio State University. https://mvapich.cse.ohio-state.edu/benchmarks/
3. Henry de Frahan, M. T., Rood, J., Day, M. S., Sitaraman, H., Yellapantula, S., Perry, B. A., Grout, R. W., Almgren, A., Zhang, W., Bell, J. B., & Chen, J. (2025). PeleC: An adaptive mesh refinement solver for compressible reacting flows. In AMReX-Combustion. GitHub. https://github.com/AMReX-Combustion/PeleC

Kestrel Containerized Benchmarks

- Containers introduced no significant penalty in scaling.
- No significant slow-down; container performance matched bare metal at large scale.

OSU[2] 3

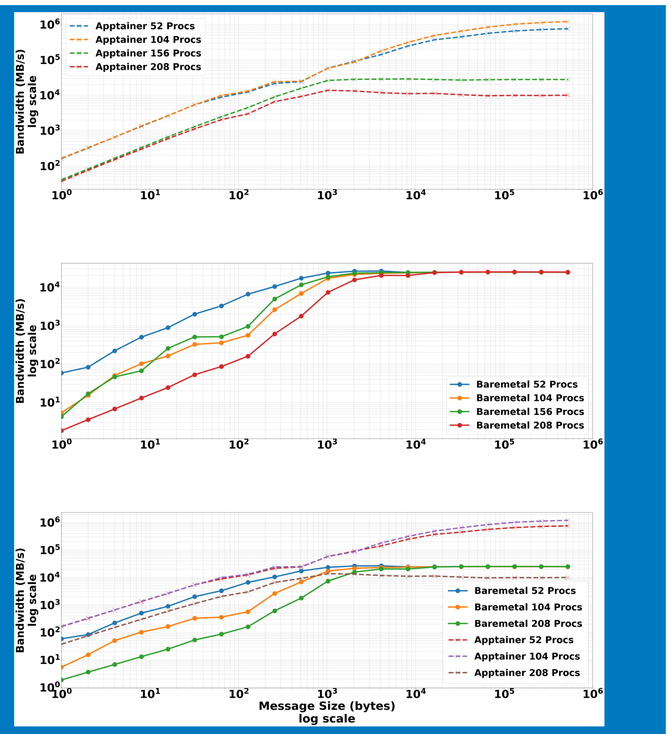


Figure 3 Multi-bandwidth graphs for multimode (4 nodes) 52-208 processors (Baremetal vs Apptainer)

- *Container bandwidth is higher for lower core counts of 52 and 104*

PeleC[3] 3

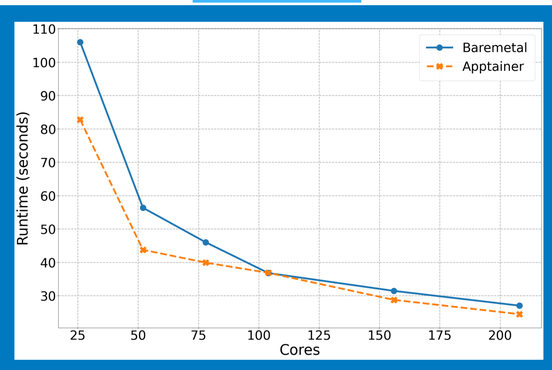


Figure 4 Strong-scaling runtime for PeleC on two node. Bare-metal and Apptainer results are shown for 26 to 208 cores.

- Apptainer ran consistently faster than bare metal across all core counts.

Acknowledgements

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